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Sub-seasonal to seasonal (S2S) prediction of dry and wet extremes for climate adaptation in India

Iqura Malik^a, Vimal Mishra^{a,b,*}^a Civil Engineering, Indian Institute of Technology Gandhinagar, India^b Earth Sciences, Indian Institute of Technology Gandhinagar, India

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ABSTRACT

Extreme climatic events have considerable impacts on society, and their prediction is an essential tool for climate change adaptation. A reliable forecast of dry and wet extremes is crucial for developing an early warning system and decision-making in agriculture and water resources. Sub-seasonal to seasonal (S2S) forecasts can be valuable for climate adaptation in water resource and agriculture sectors due to their extended range forecast ability and accessibility of different hydrometeorological products. However, the utility of these S2S models' forecasting capabilities is limited to a certain lead time, rendering them unsuitable for decision-making. We comprehensively examined the prediction skill of nine global S2S prediction models for precipitation and dry and wet extremes over India during the summer monsoon season (June to September). We find that ECCO, NCEP, and UKMO perform better than the other S2S models in predicting dry and wet extremes during the summer monsoon (June–September) in India. Our findings show that the better-performing S2S forecast models can be used to predict wet and dry extreme events several weeks ahead during the summer monsoon season. The extended range forecast system (ERFS), which is currently operational in India, provides better forecast skills for dry and wet extremes than most of the S2S models. However, S2S models provide an extended lead time forecast compared to ERFs. Therefore, a combination of ERFs and better-performing S2S models can be utilized in the early warning of dry and wet extremes at longer lead times.

Practical Implications: India has witnessed climate-related catastrophes over the past few decades that, include flooding and droughts. There is a strong need to develop tools that can provide early warning of weather and climate extremes and help in climate adaptation. Climate services and climate change adaptation need reliable forecast products at seasonal to sub-seasonal scales. Recently, sub-seasonal forecasts bridged the gap between short-range and long-range forecasts and are critical for informed decision-making in India's agricultural and disaster risk reduction sectors. We utilized S2S precipitation forecasts from various forecasting centers around the world to comprehensively examine their utility in India.

Several critical implications are associated with the findings. First, we evaluated the forecasting skill of S2S models in predicting rainfall at different regions and months of the summer monsoon season. The forecast skill of meteorological forecast varies substantially in different regions and lead times. The forecast skill weakens with the increase in forecast lead time. An improved forecast skill during the summer monsoon onset and cessation could be valuable for planning agricultural activities and water resources. In addition, we identify the regions and times where these models do not perform well and where steps can be taken to improve the model's performances in the future.

Second, there is a difference in the forecast skills of S2S models for dry and wet extremes for different regions over India. We identify a set of S2S models that provide better forecast skill for both dry and wet extremes and can be successfully employed in India's S2S operational forecast system as an early warning.

Third, we highlight the advantage of using S2S models over ERFs in forecasting dry and wet extremes in India. ERFs provide good forecast skills for both wet and dry extremes for the Indian region, but a few S2S models provide extended lead forecasts that are currently unavailable in ERFs. Therefore, we demonstrate the potential of S2S forecast information to provide early warning systems. As a result, S2S forecast information can be

* Corresponding author at: Discipline of Civil Engineering, Indian Institute of Technology Gandhinagar, India.

E-mail address: vmishra@iitgn.ac.in (V. Mishra).

integrated into a “ready-set-go” framework to provide an early warning of an extreme event a few weeks in advance.

1. Introduction

Extreme weather events, including heatwaves, cyclones, and floods, are becoming more frequent in India due to climate change. Floods, droughts, crop damage, health hazards, erosion, and water contamination are associated with extreme precipitation (Guhathakurta et al., 2011; Pall et al., 2011). Reliable precipitation forecast can help in reducing the adverse effects by providing early warning (Cloke & Pappenberger, 2009; Demirel et al., 2013; Fundel et al., 2013; Penning-Rowsell et al., 2000; Werner et al., 2006). Therefore, skillful sub-seasonal to seasonal forecast is critical for mitigating losses caused by meteorological disasters. Predictions at the sub-seasonal to seasonal (S2S) scale have recently acquired prominence due to enhanced modelling capabilities (Vitart, 2014; Zhang, 2013). An early warning of extreme events a few weeks in advance is needed; therefore, S2S prediction of extreme events might be integrated into a “ready-set-go” framework with seasonal and medium-range forecasts (Vitart & Robertson, 2018).

Wet and dry extremes have increased in many parts of the world including India during the recent decades (A. K. Mishra & Singh, 2010; Veijalainen & Vehviläinen, 2008; H. Ali et al., 2019). The severity of hydrological extremes is also projected to rise under a warming climate (Aadhar & Mishra, 2019, 2020). Jain et al. (2020) reported that India might experience droughts and floods of unprecedented magnitude in the future, surpassing recent records. In particular, the possibility of droughts with up to a 30 % deficit in the seasonal mean rainfall is a significant concern as droughts impact agriculture, water resources, and overall economic growth of the country. The frequency of concurrent hot and dry extremes is projected to increase, which is likely resulting in an increased occurrence of flash droughts under the future climate (V. Mishra et al., 2021; Kumar and Mishra, 2020). A threefold increase in widespread extreme precipitation events across central India has been observed during the 1950–2015 period (Roxy et al., 2017) that is likely to continue in the future (Mishra et al., 2022; Mukherjee et al., 2018). Extreme weather forecasting is critical and reliable meteorological forecast that can help in reducing the adverse effects of hydroclimatic extremes by providing early warning (Cloke & Pappenberger, 2009; Demirel et al., 2013; Fundel et al., 2013; Penning-Rowsell et al., 2000; Werner et al., 2006). The forecast lead time must extend to at least two weeks period to improve preparedness and minimize vulnerability to hydroclimatic extremes. Sub-seasonal to seasonal (S2S) scale prediction has recently gained prominence as a result of improved modelling capabilities (Vitart, 2014; Zhang, 2013). These extended forecasts can be used for various purposes, including providing early warnings of extreme events (drought, floods, cyclones, compound extremes) several weeks in advance, assisting in disaster preparedness, improving agricultural planning, and water resource management in India.

In recent years, the World Weather Research Program (WWRP) and the World Climate Research Program (WCRP) launched the S2S prediction project to improve sub-seasonal-to-seasonal (S2S) forecast skills (Vitart and Robertson, 2018). The S2S project provides sub-seasonal to seasonal forecasts from eleven operating centers to help in bridging forecast and knowledge gaps in developing countries (Vitart et al., 2017). The consequences of hydrologic extremes are often detrimental, including substantial economic losses (Lucas et al., 2017). Therefore, early warning systems at sub-seasonal to seasonal scales are needed for climate change adaptation. The forecast skill in short to medium term has been evaluated in India (R. Shah et al., 2017; Tiwari & Mishra, 2022); however, sub-seasonal to seasonal forecast is needed for long-term planning in agriculture and water resources. Different hydroclimatic extremes necessitate forecasting at different time scales. For

example, short to medium-term forecasts (1–2 weeks) forecast are optimal for flood early warning systems. However, seasonal forecasting can be helpful for drought mitigation as drought progresses slowly and take a longer time to develop (Butts et al., 2003). However, the predictability skills diminish with lead time, rendering meteorological/hydrological forecasts unsuitable for sector-specific decision-making (Chen et al., 2020).

Previous studies (de Andrade et al., 2019; Li and Robertson, 2015; Malik et al., 2023; Pattanaik et al., 2022; Tiwari and Mishra, 2022; Vignaud et al., 2017; Wang and Robertson, 2019; Pattanaik et al., 2020) examined the skill of S2S models in different regions. For instance, Li & Robertson (2015) evaluated the weekly prediction skill of three global prediction systems and found that the models had a decent performance in the first week. Vignaud et al. (2017) investigated the sub-seasonal precipitation forecast skill in the Northern Hemisphere summer monsoon regions at a sub-monthly lead time and reported good skills at longer than one-week lead using multimodel forecasts. On the other hand, de Andrade et al., (2019,2021) examined the sub-seasonal forecasts of global prediction systems and reported that predictability skill declines as lead time increases. Therefore, the forecast skill of meteorological forecast varies substantially in different regions and times lead time. Notwithstanding the importance of sub-seasonal to seasonal scale prediction in India due to agricultural activities and water resources planning and management, the efforts to develop a reliable forecast that can be used for climate adaptation have been limited (Ankur Paliwal, 2018; Raghu Murtugudde, 2018; Ramesh and Basu, 2019). India Meteorological Department (IMD) provides an extended range forecast system (ERFS) at the 32-day lead, which is the most-widely used product in India (Pattanaik et al., 2020, 2022; Tiwari & Mishra, 2022). S2S models provide more extended lead forecasts than ERFS. Climate services and climate change adaptation need reliable forecast products at seasonal to sub-seasonal scales. However, it remains unclear if the S2S models provide better skills than the current operational forecast from IMD. Moreover, the skill in the ensemble multi-model mean forecast of the best models from the S2S project and ERFS has not been properly examined, which could potentially improve the forecast skills for longer-lead.

2. Data and methods

2.1. Observed and forecast datasets

We used daily observed gridded precipitation data from Indian Meteorological Department [IMD] (Pai et al., 2015). The gridded precipitation data available from 1951 to 2020 was developed at 0.25° spatial resolution using station-based observations from 6,995 gage stations across India (Pai et al., 2015). The gridded dataset effectively captures precipitation variability and climatology over India during the summer monsoon (June–September) season. The gridded precipitation can represent orographic influence in the Western Ghats, foothills of the Himalayas, and north-eastern India. The observed precipitation has been widely used for hydrologic applications over the Indian sub-continent (S. A. Ali et al., 2018; V. Mishra et al., 2018; D. Shah & Mishra, 2020; Tiwari & Mishra, 2022). We evaluated the forecast skills of S2S models for the summer monsoon season, which accounts for 80 % of total annual rainfall (Jain & Kumar, 2012). We divided each month in the summer monsoon season into one to four-week leads (i.e., $d + 7$, $d + 14$, $d + 22$, $d + 28$), where d refers to the starting date of the model reforecast obtained from the S2S database. We used observed IMD precipitation corresponding to each model’s available retrospective forecast for comparison.

Table 1

Summary of S2S forecast products used in this study.

Models	Time range	Resolution	Reforecast size	Reforecast length	Frequency
CMA	0–60	T266 L56	4	past 15 years	2/week (Mon, Thu)
ECCC	0–32	39 km/ L45	4	1998–2017	Weekly (Thu)
ECMWF	0–46	Tco639/319 L91	11	past 20 years	2/week (Mon, Thu)
HMCR	0–61	1.1° x 1.4° L28	11	1991–2015	Weekly (Thu)
ISAC	0–32	0.75° x 0.56° L54	5	1999–2018	every 5 days
JMA	0–33	T1479/T1319 L100	5	1991–2020	2/month
KMA	0–60	N216 L85	7	1991–2016	4/month
NCEP	0–44	T126 L64	4	1999–2010	Daily
UKMO	0–60	N216 L85	7	1993–2016	4/month

Reforecast is a set of forecasts with historical start and projection dates, which can be used to examine the forecast skill against the observations. We used the S2S database which consists of reforecasts and near-real-time forecasts from eleven global prediction centres (Vitart et al., 2017). We obtained data from nine global prediction centers that provide a consistent forecast, which include China Meteorological Administration (CMA), Korea Meteorological Administration (KMA), Hydro meteorological Centre of Russia (HMCR), Japan Meteorological Agency (JMA), Environment and Climate Change Canada (ECCC), ECMWF, the Met Office (UKMO), National Centres for Environmental Prediction (NCEP), and the Institute of Atmospheric Sciences and Climate (ISAC) (Table 1).

We regridded the reforecasts from the selected models to 0.25° using bilinear interpolation to match the resolution of the observed gridded precipitation products. Some of the S2S models (CMA and NCEP) have fixed reforecasts, as these produce reforecasts once during the lifetime of a given model version. On the other hand, on-the-fly configuration reforecasts are produced simultaneously as real-time forecasts (as for ECMWF, KMA, and UKMO). Also, the on-the-fly configuration is considerably more computationally demanding when it comes to generating updated reforecasts in real-time (Peng et al., 2023). In addition, the beginning period, reforecast frequency, and length of forecast vary in each model. We determined the largest time period for which all the models provide a forecast. Furthermore, all the models have different forecast lead times, but all of them have a minimum duration of 28 days. Therefore, we opted a four-week lead time for our analysis.

Each model contains a control reforecast member that employs a single unperturbed initial condition and perturbed forecast members that account for uncertainty in the initial and boundary conditions. We used the mean of all the ensemble members (control and perturbed) to obtain a deterministic forecast that can be compared with the observed data. In order to evaluate model performance, we collected the maximum number of years of data that each model can provide during the summer monsoon season (Table S1). For each month in the monsoon season, we obtained data twice a month: once at the earliest initial date possible and once in the middle of the month, which resulted in eight samples for each model during the summer monsoon season. We prepared observed data from IMD by matching the dates with each of the S2S models to evaluate the forecast skill using different verification metrics.

Apart from S2S models, we used meteorological forecasts from the four models that participated in Extended Range Forecast System (ERFS) to evaluate dry and wet extremes in India. The ERFS model was developed at the Indian Institute of Tropical Meteorology (IITM), Pune, and implemented by IMD to provide operational Extended Range Forecast

products to different users (Abhilash et al., 2014; Pattanaik and Das, 2015; Shah et al., 2017; Vegad and Mishra, 2022). The four models that are currently operational in ERFS forecast are: (i) CFSv2 at T382 (≈ 38 km), (ii) CFSv2 at T126 (≈ 201 100 km), (iii) GFSbc (bias-corrected SST from CFSv2) at T382, and (iv) GFSbc at T126. T382 202 and T126 represent around 38 km and 100 km of spatial resolution, respectively. Atmospheric and oceanic initial conditions are available from the National Center for Medium-Range Weather Forecasting (NCMRWF) and Indian National Centre for Ocean Information System (INCOIS), respectively. The multimodel ensemble (MME) of the four ERFS models is operational at 32 days lead. The same suites of the model are also run on hindcast mode, available for 16 years (2003 to 2018). We used the control forecast of all four models. These models are at different resolutions; therefore, we regridded them at 0.25-degree resolution to match them with IMD precipitation and the S2S forecast products.

2.2. Evaluation of precipitation forecast

We obtained summer monsoon season precipitation from all nine S2S models and examined their performance using three deterministic forecast verification measures. As no single verification metric captures all aspects of forecast quality (Murphy, 1993), it is essential to evaluate forecast skills using a variety of statistical measures. Therefore, we used percentage bias (PBIAS), root-mean-square error (RMSE), and anomaly correlation coefficient (r) to compare observations to the ensemble mean of all reforecast members (control plus perturbed members) from a forecast model. In addition, we used an additional score to evaluate and rank each model in terms of their forecast skill, which includes different attributes of verification to evaluate the overall performance of each model. The overall performance score was used (Graham et al., 2019; Mahto & Mishra, 2019), which was calculated as follows:

$$\text{Score} = [(r) + (1 - (\text{abs}(\text{bias})/\text{Range})) + (1 - \text{RMSE}/\text{Range})] \times 10/3 \quad (1)$$

where *Range* is the difference between the maximum and minimum value of rainfall in the monsoon season during a common time period for all models. We utilized a deterministic score to evaluate the model's performance, which is correlated with the model's probabilistic skills (Sooraj et al., 2012; Alessandri et al., 2011; Yang et al., 2017). Additionally, estimating deterministic forecast skill is computationally less demanding and easier to interpret compared to the probabilistic forecast skill. However, the probabilistic forecast skill provides uncertainty estimates of different forecast ensemble members that can be examined in the future (Nanditha and Mishra, 2020).

2.3. Forecast skill of hydroclimatic extremes

We evaluated the forecast skill of each S2S model in predicting both dry and wet extreme events. To identify each extreme event, we selected the period 1999–2010 for evaluation, as it was the common period for all the forecast products. We used the Standardized Precipitation Index (SPI; McKee et al., 1993) to identify dry and wet extremes. Weekly SPI was calculated using the weekly rainfall total after fitting the Gamma distribution (Aadhar & Mishra, 2017). We used the Drought Severity and Coverage Index (DSCI; Leelaruban et al., 2012) to examine the severity and coverage of dry extremes during the summer monsoon season, which can be estimated as:

$$\text{DSCI} = 1(D0) + 2(D1) + 3(D2) + 4(D3) + 5(D4) \quad (2)$$

Here, D0, D1, D2, D3, and D4 represent the area (%) of abnormal drought, moderate drought, severe drought, extreme drought, and exceptional drought, respectively. DSCI ranges from 0 to 500. Zero means that none of the areas is abnormally dry or under drought, while 500 means that all the area is under the D4 (exceptional) drought category. Like drought, we determined the severity and coverage of wet extremes using the Wet Severity and Coverage Index (WSCSI). The per-

Table 2
Summary of categorical verification measures used in the study.

Criterion	Formula	Range	Reference
Hit Rate (HR)	Hits / (Hits + Misses)	0–1	(Swets, 1986)
False Alarm Rate (FAR)	False rate / (False rate + Hits)	0–1	(Swets, 1986)
Critical Success Index (CSI)	Hits / (Hits + Misses + False rate)	0–1	(Gilbert, 1884)

centage area of the abnormal wet, moderate wet, severe wet, extreme wet event, and exceptional wet events is represented by W0, W1, W2, W3, and W4, respectively. The WSCI scale ranges from 0 to 500. Zero indicates that no part of the area is abnormally wet, while 500 indicates that the entire area is in W4, the exceptional wet category.

$$WSCI = 1(W0) + 2(W1) + 3(W2) + 4(W3) + 5(W4) \quad (3)$$

A few S2S models (ECCC, ISAC, and JMA) provide reforecasts only for up to one month; forecast skill for these models was estimated for three weeks lead. Other models (CMA, ECMWF, KMA, NCEP, HMCRC, and UKMO) provide forecasts of up to five weeks of lead time.

For assessing the quality of the forecast, both skill and value of the forecast are important. The skill provides us with an idea of how well the forecast predicts the observed conditions, while forecast value tells us whether the forecast outperforms climatology or not (Dorninger et al., 2018). We evaluated the forecast skills of the models using categorical skill verification scores, including hit rate (HR), false alarm rate (FAR), and critical success index (CSI) (Table 2). A perfect score for both (HR and CSI) skill scores is one and zero for FAR. The CSI is more complete than the HR and FAR as it can detect both missed events and false alarms (Chen & Wilson, 2014). We also estimated forecast value, which is based on a perfect forecast system and is calculated using skill score (Jolliffe & Stephenson, 2012).

$$\text{Skill Score} = \frac{\text{Score of forecast} - \text{Score of the standard forecast}}{\text{Perfect score} - \text{Score of Standard forecast}} \quad (4)$$

A score greater than zero indicates that the forecast adds value to climatology, whereas a score less than zero suggests that climatology performs better than the forecast. Here we used CSI as a score of forecasts with a perfect score of CSI as 1.

3. Results

3.1. Precipitation forecast skill

First, we examined the spatial distribution of rainfall climatology for each month of the summer monsoon season for the period 1980–2019 to

determine the rainfall forecast skill of each model (Fig. 1). During June, the maximum rainfall is observed over the west coast and north-eastern regions of India. In July, a marked increase in rainfall occurs throughout the region, especially in central India. July and August contribute more to total summer monsoon precipitation than June and September (Fig. 1).

We used three deterministic verification metrics (PBias, correlation coefficient, and RMSE) to evaluate each S2S model over the Indian region. We estimated PBias and anomaly correlation for rainfall from the S2S models and observations (IMD) over India for lead times of up to four weeks during the summer monsoon season (Fig. 2a and 2b). The majority of S2S models, including ECCC, ECMWF, HMCRC, ISAC, KMA, NCEP, and UKMO, exhibit a strong correlation with observed precipitation at a lead time of one week. On the other hand, JMA and CMA show weak correlation, which declines as the lead time increases. JMA, ECCC, and UKMO display relatively lower bias at a one-week lead. Bias in precipitation forecast increases with lead time in all the S2S models. KMA and HMCRC overestimate precipitation at all lead times in all the weeks and across the regions. NCEP, ECMWF, and ISAC, on the other hand, underestimated precipitation in several regions over India. The rest of the S2S models exhibit a mixed signal with higher spatial variations in the forecast skills. Most of the S2S models show a higher forecast skill at one-week lead, while a majority of them overestimate rainfall in the fourth week. In northwest India, a majority of the S2S models overestimate precipitation at a 3–4 week lead. On the other hand, most of the S2S models underestimate precipitation in peninsular India at longer leads. Overall, we find that despite a good correlation between observed precipitation and forecast from the S2S models, the bias in the forecast is substantial at all lead times. Moreover, both dry and wet bias in precipitation forecast from the S2S models can pose challenges for the decision-making related to agriculture and water resources.

We further examined the temporal distribution of model biases for different months and lead times over India during the summer monsoon season (Fig. 3). We estimated all-India median bias from all the S2S models for June, July, August, and September. We find that ISAC, JMA, and KMA show a large bias, whereas ECCC, ECMWF, NCEP, and UKMO exhibit relatively lesser bias in the precipitation forecast in June. Most of the S2S models show relatively lesser bias and better precipitation forecast skills in July compared to June (Fig. 3a). ECCC, ECMWF, and HMCRC provide lower bias in July, consistent to June forecast. On the other hand, CMA, ISAC, and JMA overestimate rainfall in July; however, the wet bias is lesser than the bias in the forecast for June. Precipitation forecast skill further improves in a majority of S2S models for August than in July. ISAC and JMA overestimate precipitation in all the weeks of August, as they did in July. This could be because the two models have identical configurations, such as fixed reforecast configuration and atmosphere-only models (Endris et al., 2021). In September, except for

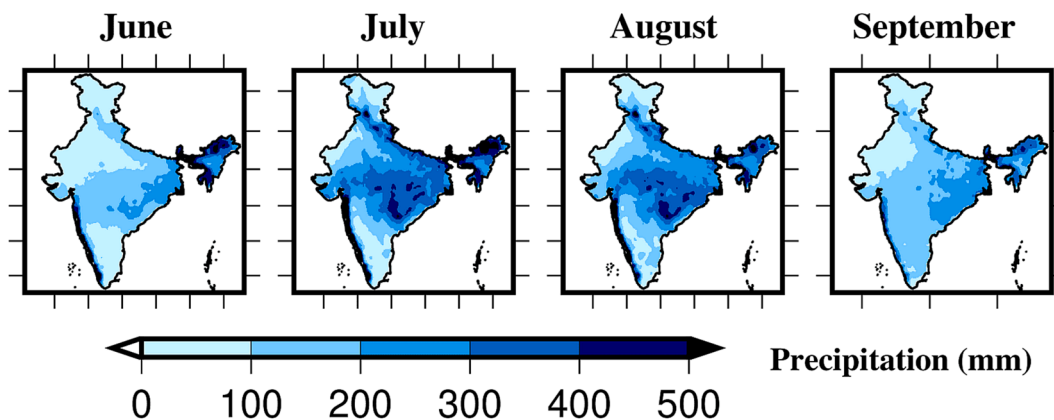


Fig. 1. Spatial pattern of climatological mean precipitation for each month (June, July, August and September) of the Indian Summer Monsoon season (JJAS) Precipitation for 1980–2019 period.

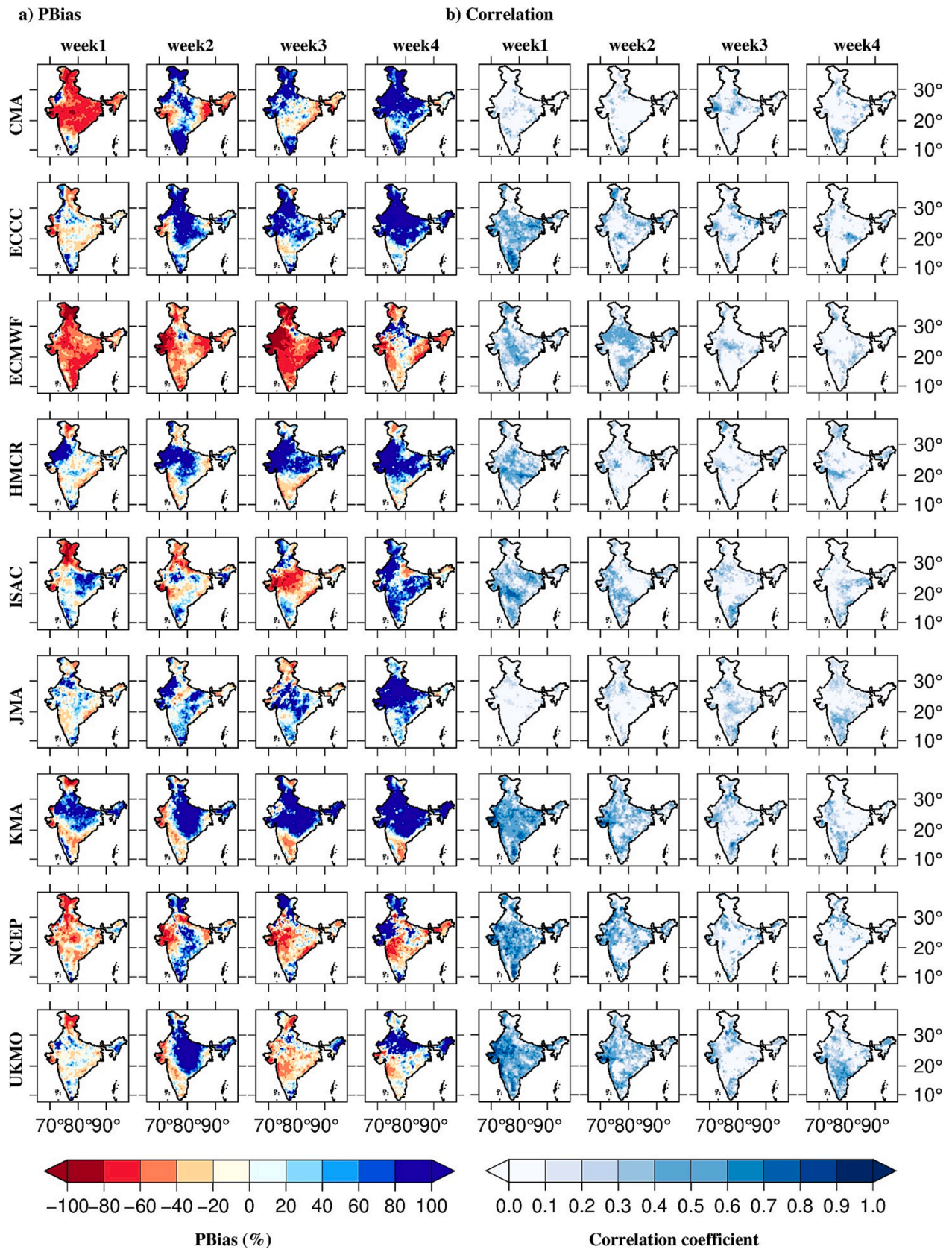


Fig. 2. Performance evaluation of S2S models on a weekly basis over Indian region for a common time frame of 1999–2010. Spatial distribution of (a) PBias and (b) Anomaly correlation coefficient between S2S model's and observed rainfall (IMD) during the summer monsoon (JJAS) over India. Blue shade in (a) shows overestimation while the red shade shows an underestimation of precipitation from observations.

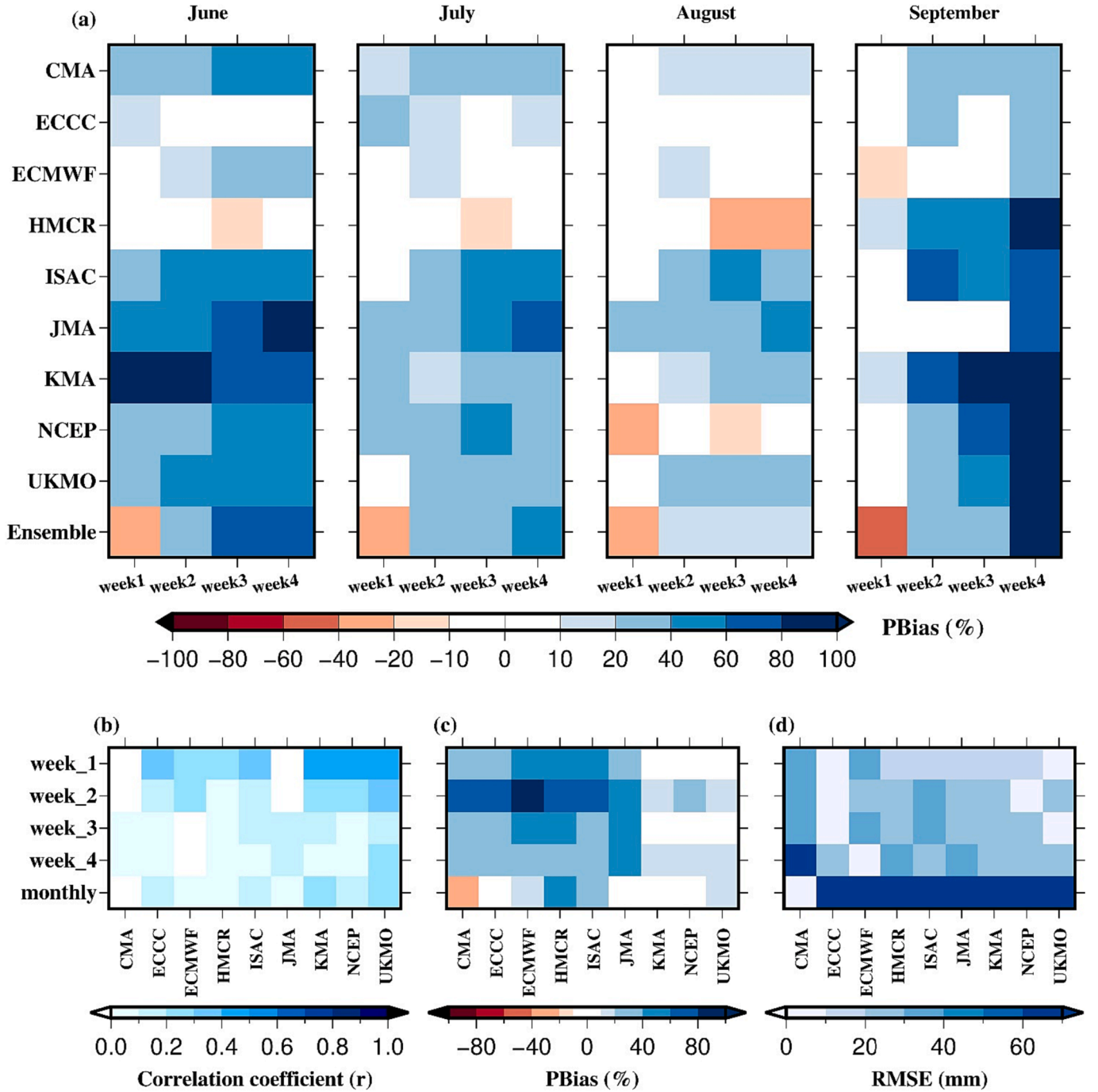


Fig. 3. Performance evaluation of S2S models at different time scales (monthly and weekly) for a common period of 1999–2010. (a) Month-wise variation in PBias in Indian monsoon rainfall between S2S models and IMD observations for each month of monsoon season. Blue shades represent the overestimation of the model and brown shades represent the underestimation of model over observations, (b) week-wise variation of the anomaly correlation coefficient, (c) PBias and (d) RMSE between each S2S model and observed IMD precipitation during monsoon season.

the week one lead, most of the S2S models overestimate precipitation.

We find that the majority of the S2S models show less bias during the peak of the summer monsoon, which occurs in July and August. Whereas, during the onset (June) and cessation (September) of the summer monsoon, a large bias, and low forecast skill was noted (Fig. 3a). In June and September, almost all the models exhibited large bias, which increases with lead time. In contrast, forecast skill is better at all the leads during the peak summer monsoon season (July and August). In addition to forecast skill at different times during the summer monsoon season, we also examined the spatial variability in the skill during the summer monsoon season (Fig. S1). The highest spatial

variability in bias (Fig. S1) is found in HMCR, followed by ISAC. The CMA model showed the least variation for week one; however, spatial variability in bias increases with the lead time. In contrast, ECMWF, ECC, JMA, KMA, NCEP, and UKMO showed less variability in bias across India. We find that the forecast skill of HMCR and ISAC vary from one region to another, whereas ECC, ECMWF, and NCEP show less variability in bias in all the regions. Our findings show that the ECC, KMA, UKMO, NCEP, HMCR, and ECMWF exhibit a strong correlation, while the CMA and JMA models display a weak correlation at a lead time of one week (Fig. 3b), which decreases with the increase in lead time. The UKMO model maintains a good correlation across all the lead times.

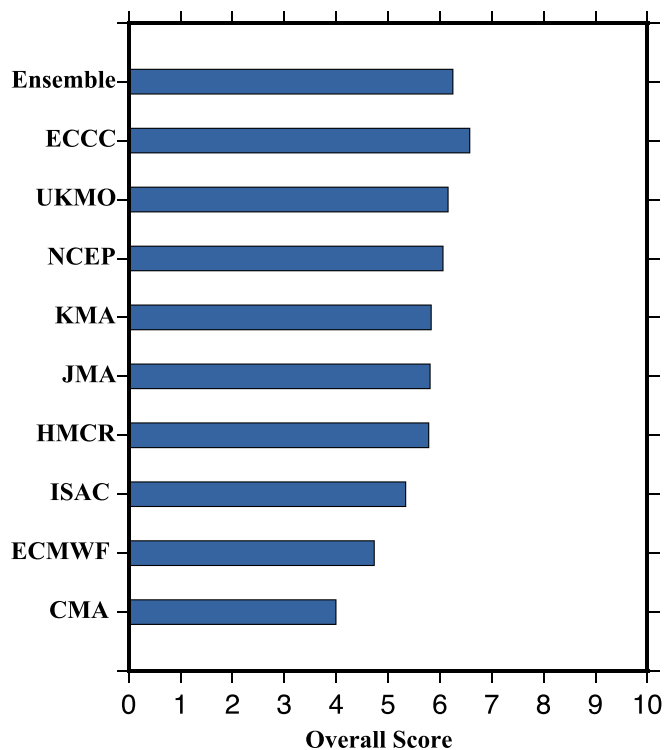


Fig. 4. Performance Score (Out of 10) for the S2S model's total precipitation predictability skill in India. Higher values show good performance, and low values show lower performance of models. The ensemble score refers to the ensemble of three top-performing models (ECCC, UKMO, and NCEP).

Similar performance was noted based on bias (Fig. 3c) and RMSE (Fig. 3d) during the summer monsoon season.

Based on the overall score (Table S2), we find that the ECCC model outperforms the other S2S models for precipitation forecast, while the UKMO ranks second among the nine S2S models (Fig. 4). The other S2S models including NCEP, KMA, and JMA show a good performance for precipitation forecast in the summer monsoon season. On the other hand, ECMWF, ISAC, HMCR, and CMA showed weaker performance in the Indian region. We also evaluated the performance of the ensemble of the top three performing models (ECCC, NCEP, and UKMO) against the individual models. The combined score (6.25) of the ensemble from the better-performing models is higher than most models (except for ECCC). [DelSole and Shukla \(2012\)](#) used forecasts from the UKMO and ECMWF models for predicting the Indian summer monsoon at the seasonal scale and reported superior performance of UKMO compared to the other models. Similarly, [Jain et al. \(2018\)](#) showed that the multimodel ensemble outperformed the individual models, which is consistent with our findings.

3.2. Forecast skill of hydroclimatic extreme events

Next, we evaluated individual S2S models for their ability to predict dry and wet extremes. We used observed precipitation from IMD and forecast from ERFs and S2S models to identify dry and wet extremes using SPI during the summer monsoon for the 2003–2010 period. Since the precipitation forecast from different models is available for different times, a common period in all the models was selected for the analysis. Using SPI from observed precipitation and S2S forecast, we reconstructed the wet and dry events to estimate at least three-week lead forecast skills of the S2S models. We calculated DSCI and WSCI to compare the observed and predicted wet and dry extremes. We identified July 01–07, 2007 as the wettest period while August 01–07, 2009 as the driest period based on weekly data (Fig. S2).

While the majority of models provide a five-week lead, the three S2S models (ECCC, ISAC, JMA) and ERFs do not provide forecast of up to five-week lead. We find that ECCC shows a better forecast skill than the other models at a three-week lead time for wet extreme (Fig. 5). However, the ECCC fails to predict the wet extreme centered over western India at week 1 and week 3 leads. ISAC shows higher forecast skill for WSCI at a one-week lead but fails to predict wet extremes at three weeks lead. JMA, on the other hand, overestimates the wet extremes extent at all lead times. ERFs perform well at one week lead time; however, the ERFs model did not capture the severity of wet extremes over western India at one and three weeks lead. Wet extremes on the west coast are captured by all the forecast models, including ERFs, regardless of their lead times. Only NCEP, UKMO, and KMA (Fig. 6) show reasonable forecast skill at one week lead time. ECMWF overestimates the wet extreme extent at one week lead time. In most regions, the spatial distribution of SPI from KMA and UKMO is nearly identical, which could be due to the same model configuration. However, different models have different atmospheric initial conditions ([Noh et al., 2016](#)).

We identified August first week of 2009 to be the driest event (Fig. S2). ISAC, ERFs, UKMO and CMA provide good forecast skills for dry extremes at one to three-week lead time. In contrast, NCEP exhibits medium forecast skill at the lead times. All the other models (Figs. 7 & 8) fail to capture the dry extreme event successfully and underestimate the severity of the dry extreme.

3.3. Forecast skill for dry and wet extremes in the Krishna river basin

After a severe drought in the summer monsoon season of 2009, the Krishna river basin experienced a devastating flood in late September 2009. Several districts in Karnataka and Andhra Pradesh were inundated by unprecedented rainfall in North Karnataka. Fourteen states and 338 districts in India witnessed drought-like conditions in 2009 ([Ministry of Agriculture, 2009](#)). The drought of 2009 was one of the deadliest droughts in India ([Richard Mahapatra, 2001](#)). Considering the temporal variability of rainfall extremes during the summer monsoon, we examined forecast skills of dry and wet extremes three weeks in advance using the three better-performing S2S models (UKMO, ECC, and NCEP) and the currently operational ERFs for the extended range forecast.

We analyzed the extreme drought conditions over the Krishna basin in the first week of August 2009 (Fig. 9). All three S2S models, with the exception of ECCC, predicted dry weather at a lead time of one week. We also examined if a multimodel ensemble of these four models provides better forecast skill. To better determine spatial coverage and intensity of drought, we used DSCI as drought severity and coverage index. Only ERFs and NCEP showed better forecast skills at the three-week lead. Due to the weaker performances of ECCC model, ensemble mean of the models' forecasts did not provide a good forecast skill.

In late September, the basin experienced an extreme wet condition following a prolonged drought. ERFs, NCEP, and UKMO show reasonable forecast skill at lead times of one and three weeks (Fig. 10). Additionally, these models fairly depict the percentage of wet areas using WSCI. ECCC gives a moderate skill of forecast for the wet event compared to extreme dry condition. All four models and their multimodel ensemble provide good forecasting skill at a one-week lead. However, only a few models (UKMO and multimodel ensemble) provide reasonable skill beyond one-week lead time. Therefore, the Indian operational forecasting system can effectively use the ERFs, UKMO, and NCEP models for forecasting wet and dry extremes and provide early warning.

We estimated performance of each model using categorical skill scores for eight events (details of which are given in Table S3) of dry and wet extremes during the 2003–2010 period based on SPI (all India average) to further examine the robustness of our results. We estimated hit rate, false alarm, and critical success index (CSI) for the three best forecast models (Fig. 11) for both dry and wet extremes. A high hit rate and a low false alarm rate is observed for the three best-performing S2S

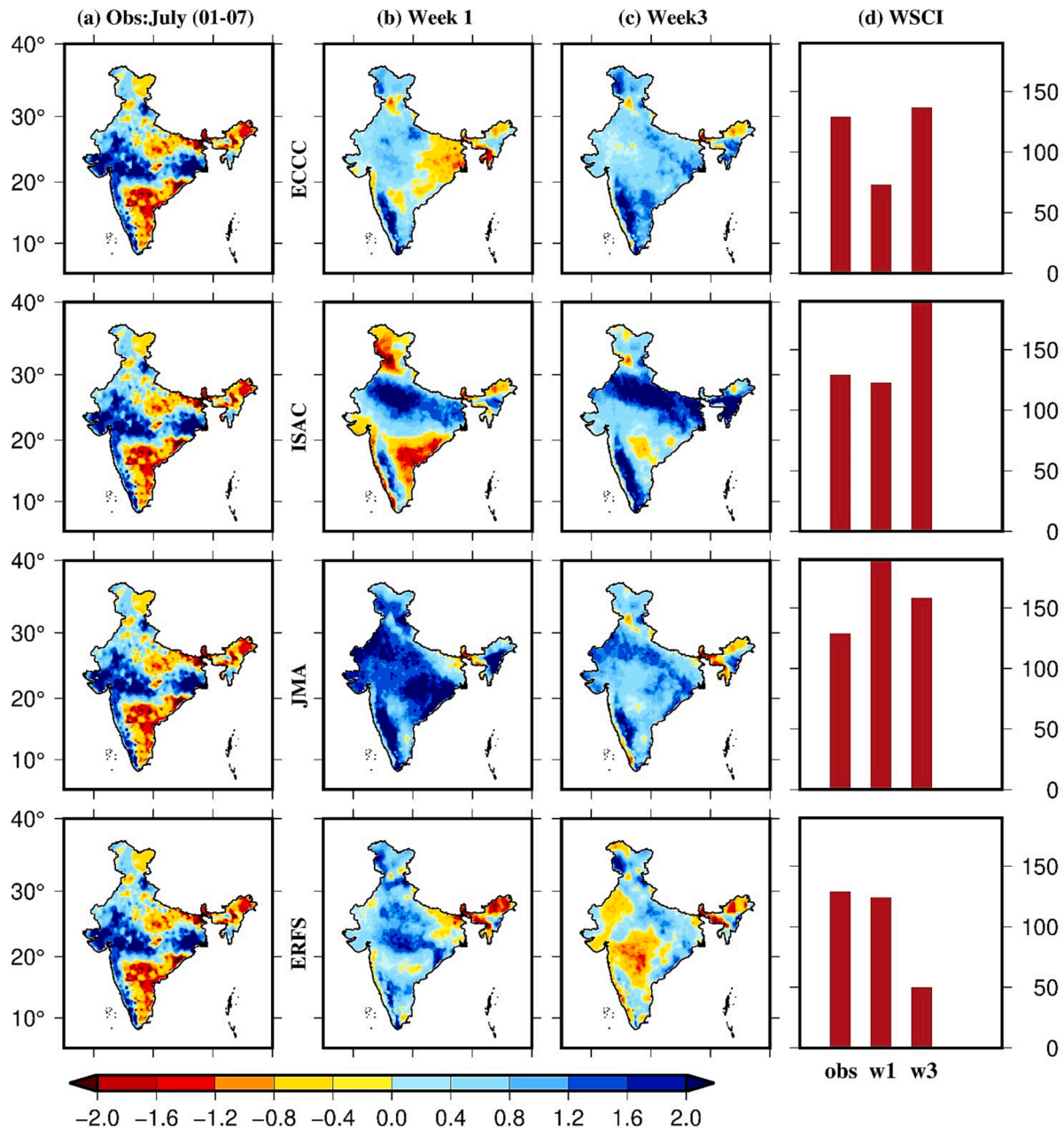


Fig. 5. (a–c) Comparison of wet extreme based on (a) observed SPI against S2S model initialized (b) one week and (c) three weeks before extreme wet event of 2007. (d) WSCI of wet event from observed data (obs) and from models initialised 1 week (w1) and 3 weeks (w3) before of the wet event respectively.

models and ERFs. A decline in hit rate with the increase in lead time was found for most of the models. ERFs ranks first in predicting the dry extremes in terms of hit rate and provides a good forecast skill for both wet and dry extremes. Overall, our results show that UKMO and ERFs perform better among the S2S models. Overall, all the four (ECCC, NCEP, UKMO, and ERFs) forecast models demonstrate the potential for sub-seasonal to seasonal hydroclimatic prediction, which can aid in the decision-making of India's water management and agricultural activities. We also estimated CSI, which measures the fraction of correctly forecasted events to the total number of events. All the models (UKMO ECCC, NCEP and ERFs) show positive score values at all the lead times, indicating that the models aid value over the climatology and can be successfully used in operational forecasts in India (Fig. 11).

4. Discussion and conclusions

We evaluated the ability of nine state-of-the-art operational models from the S2S database to predict precipitation and hydroclimatic extremes during the summer monsoon season. We used different deterministic and categorical verification metrics to assess the skill using the ensemble mean of all forecast products. Precipitation forecast skill varies in space and time in different S2S models during the summer monsoon season.

Most of the S2S models underestimate orographic precipitation which is difficult to model and observations over high topography have large uncertainties due to sparse observational networks (Pokhrel et al., 2016). Moreover, the spatial resolution of the S2S models may be too coarse to resolve orographic precipitation. Spatial variability in the performance of the models is also evaluated to examine if models give a similar performance at all locations, which can help forecasters to

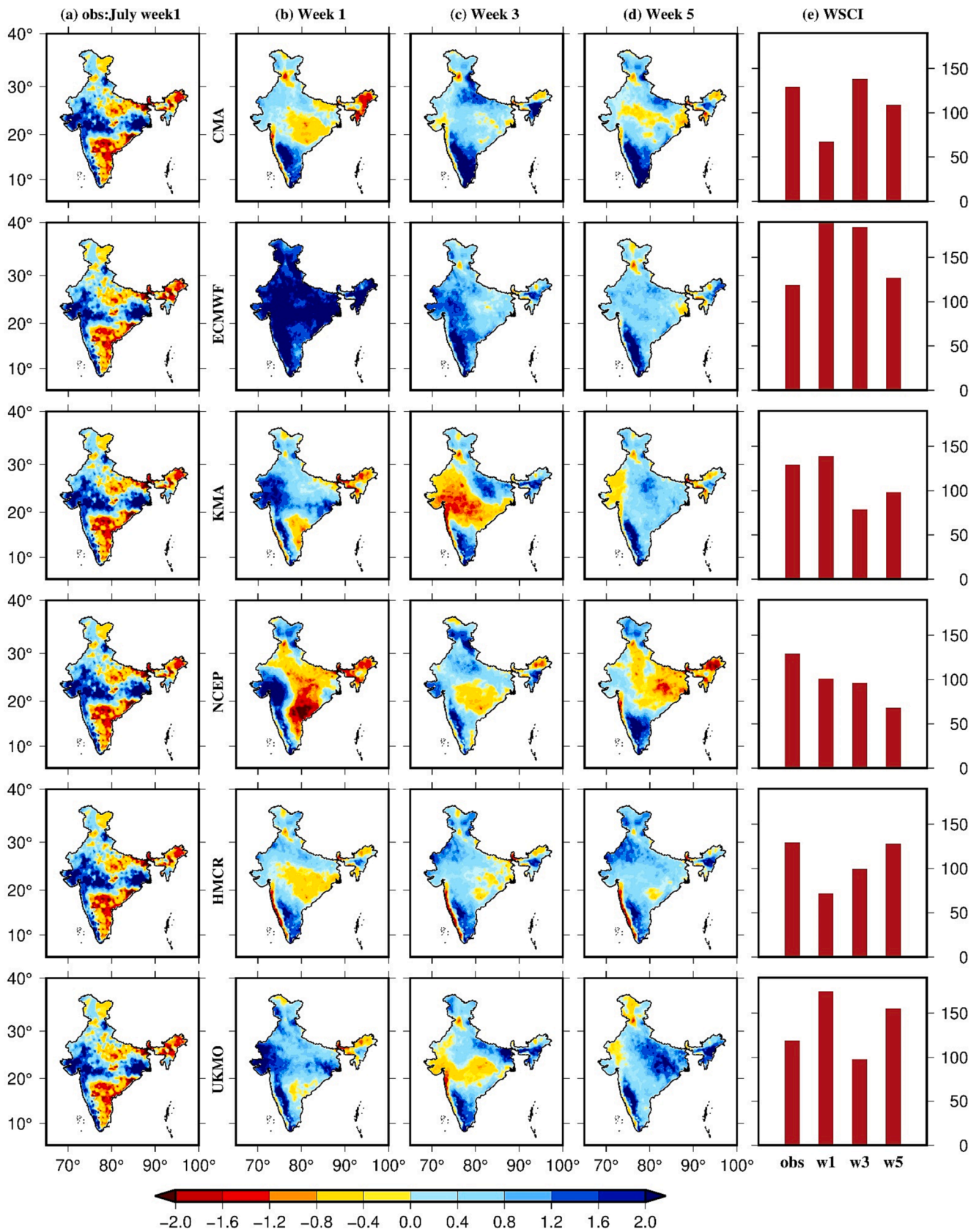


Fig. 6. (a–d) Comparison of peak wet based on (a) observed SPI against S2S model initialized (b) one week, (c) three weeks and (d) five weeks before extreme wet event of 2007. (e) WSCI of wet event from observed data (obs) and from models initialised 1 week (w1), 3 weeks (w3) and 5 weeks (w5) before of the wet event respectively.

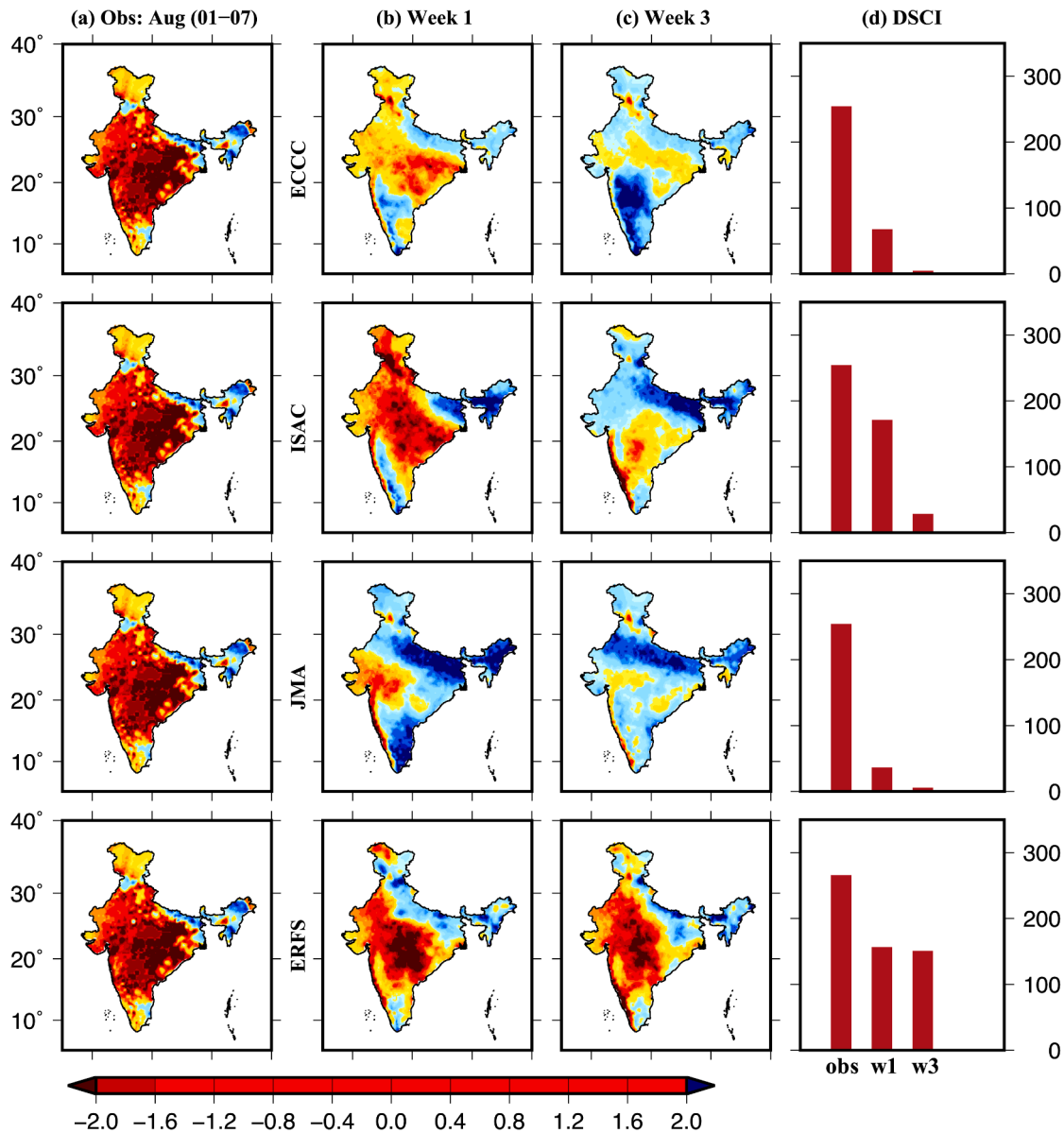


Fig. 7. (a–c) Comparison of peak dry event based on (a) observed SPI against S2S model initialized (b) one week and (c) three weeks before dry extreme event of 2009. (d) DSCI of dry extreme event from observed data (obs) and from models initialised 1 week (w1) and 3 weeks (w3) before of the dry event respectively.

choose models that are consistent in their performance irrespective of the location.

The models generally perform well during the peak of the rainy season (i.e., July and August) but poorly during the onset and termination stages (June and September) of the summer monsoon season. Predicting summer monsoon onset and cessation is complex as it is linked with local synoptic and large-scale processes like ENSO, IOD, and MJO. The MJO dominates intraseasonal convective activity in the tropics, with a significant impact in the intermediate and high latitudes. At the sub-seasonal time scale, MJO is regarded as a major source of global predictability (Lau & Waliser, 2012). UKMO model has a higher bivariate correlation for the MJO than the other models; this could be the reason for its superior performance at all the leads. However, S2S models have difficulty in capturing MJO propagation in western-pacific countries (Vitart, 2017). Therefore, some S2S models may fail to predict the intensity of monsoon intra-seasonal oscillations over the Arabian sea, which are stronger in June and September (Chattopadhyay et al., 2018). The poor prediction skill in June could also be attributed to a progressive directional shift in low-level winds as easterly jets attempt to

establish themselves to activate the full force of the southwest summer monsoon (O'hare, 1997; Y.P. Rao, 1976).

Forecast skills in the S2S models deteriorates as lead time progresses. ECCC, KMA, NCEP, and UKMO perform better than the others. In terms of the overall score, ECCC, NCEP, and UKMO outperform the other S2S models. The best and poor-performing models are consistent with findings from recent studies (de Andrade et al., 2019; Endris et al., 2021; Zhou et al., 2019). The better-performing models generally better capture large-scale drivers like the MJO (Endris et al., 2021). The model's performance is also strongly influenced by its initial state and model resolution (Zhou et al., 2019), which can contribute to better forecast skills of the three better-performing models.

Extreme dry and wet events are projected to become more frequent in the future. We evaluated the ability S2S models to predict dry and wet extremes at different lead times ranging from one to four weeks in India using S2S precipitation forecast products during the summer monsoon season.

Based on our findings, we draw the following conclusions:

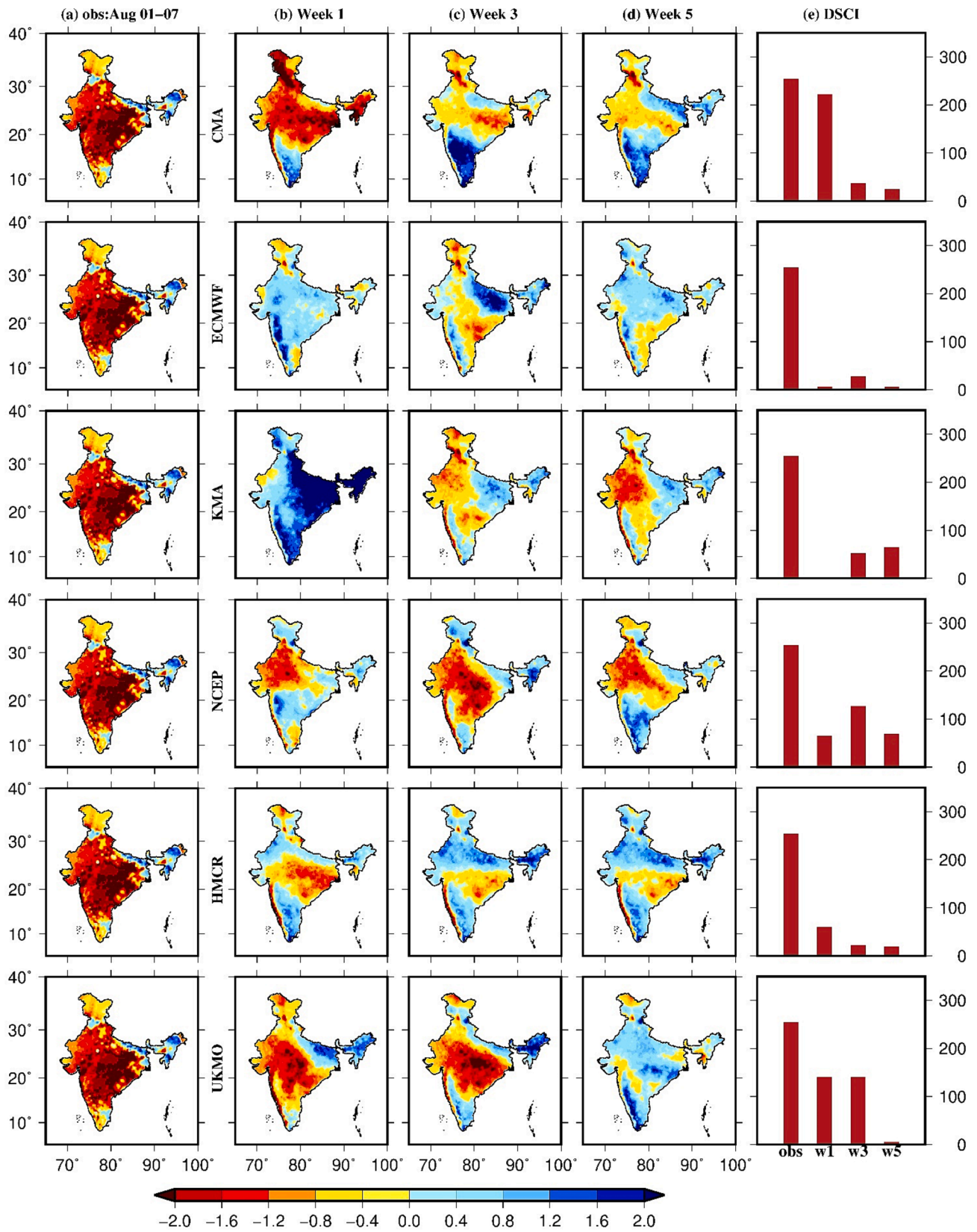


Fig. 8. (a–c) Comparison of peak dry event based on (a) observed SPI against S2S model initialized (b) one week, (c) three weeks and (d) five weeks before dry extreme event of 2009. (e) DSCI of dry extreme event from observed data (obs) and from models initialised 1 week (w1), 3 weeks (w3), and 5 weeks (w5) before of the dry event respectively.

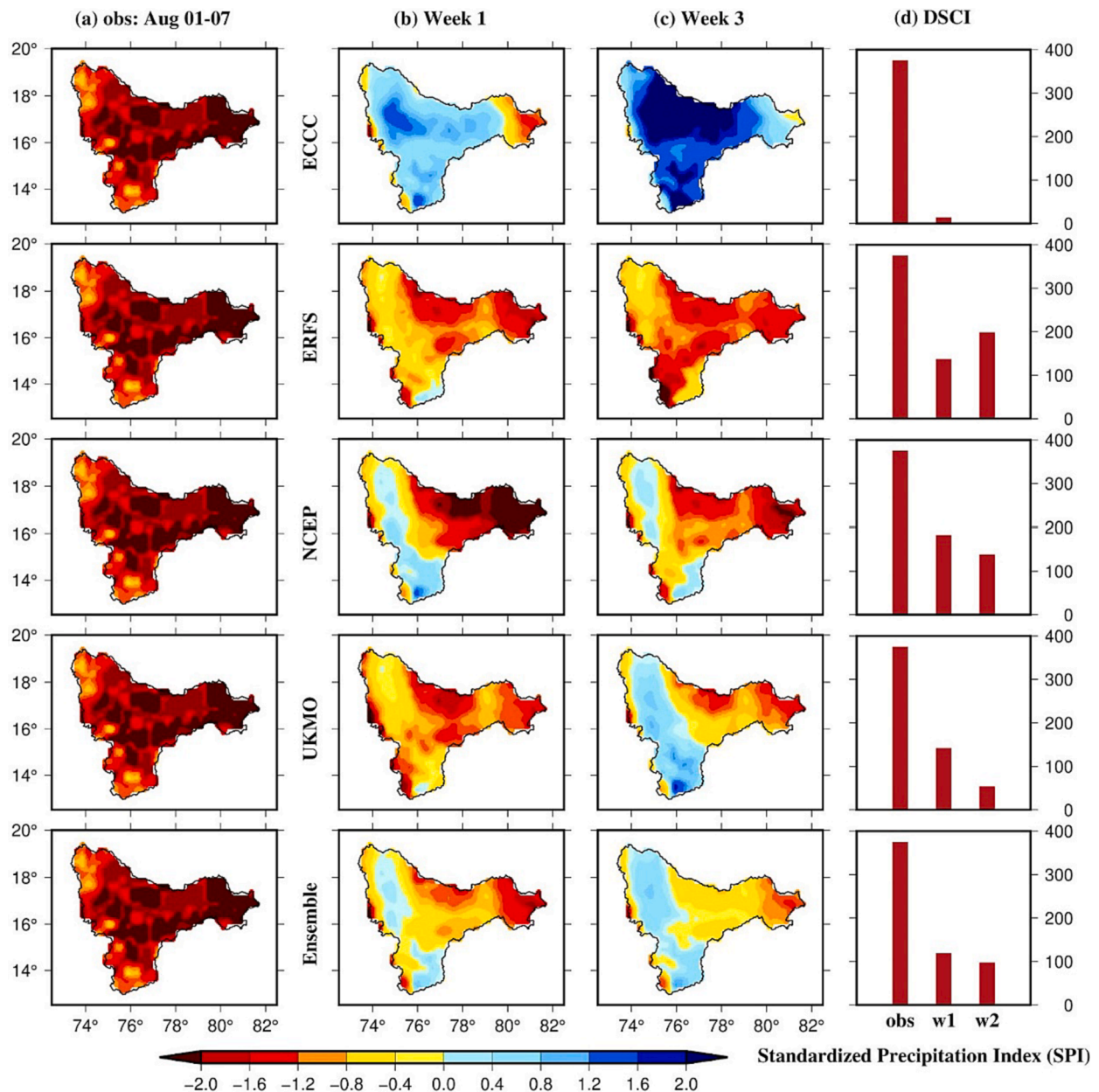


Fig. 9. (a–c) Comparison of Aug 2009 dry event based on (a) observed SPI against S2S model initialized (b) one week, (c) three weeks before dry extreme event, (d) DSCI of dry extreme event from observed data (obs) and from models initialised 1 week (w1) and 3 weeks (w3) before of the dry event respectively.

- The precipitation forecast skill of S2S models varies both geographically and temporally, depending on the initialization week. The forecast skill weakens with the increase in forecast lead time. The best precipitation forecast skill from the S2S models is shown in July and August, which is the peak of the summer monsoon season. Almost all the S2S models show a relatively weaker forecast skill for the onset (June) and termination (September) of the summer monsoon. An improved forecast skill during the summer monsoon onset could be valuable for the planning of agricultural activities and water resources.
- For both dry and wet extremes, the S2S models UKMO, NCEP, and ECCC perform better than others. These models (UKMO, NCEP, and ECCC) perform better in predicting wet extremes at a three-week lead compared to dry extremes. In addition, these S2S models were successful in forecasting the extreme wet event after a prolonged dry period in the Krishna River basin in 2009, indicating that the S2S

models can be successfully applied in India's S2S operational hydrological forecast system as an early warning.

- We compared the forecasting skills of S2S models against ERFS, which is an operational forecast in India. ERFS provide good forecast skill for both wet and dry extremes compared to a majority of S2S models. Forecast skill requirements vary based on the application, with critical purposes like disaster management and agriculture planning demanding higher forecast skills. For forecasts beyond 32 days, S2S forecasts can be valuable in extending IMD's forecast horizon. Moreover, large ensemble members available from the S2S models can assist in quantifying forecast uncertainty and development of probabilistic forecast of meteorological and hydrological variables. Our findings can be valuable in developing an integrated early warning system combining the existing operational forecast from IMD and S2S.
- Our research focused primarily on the Indian summer monsoon season. In future, S2S forecasts of extreme events could be integrated

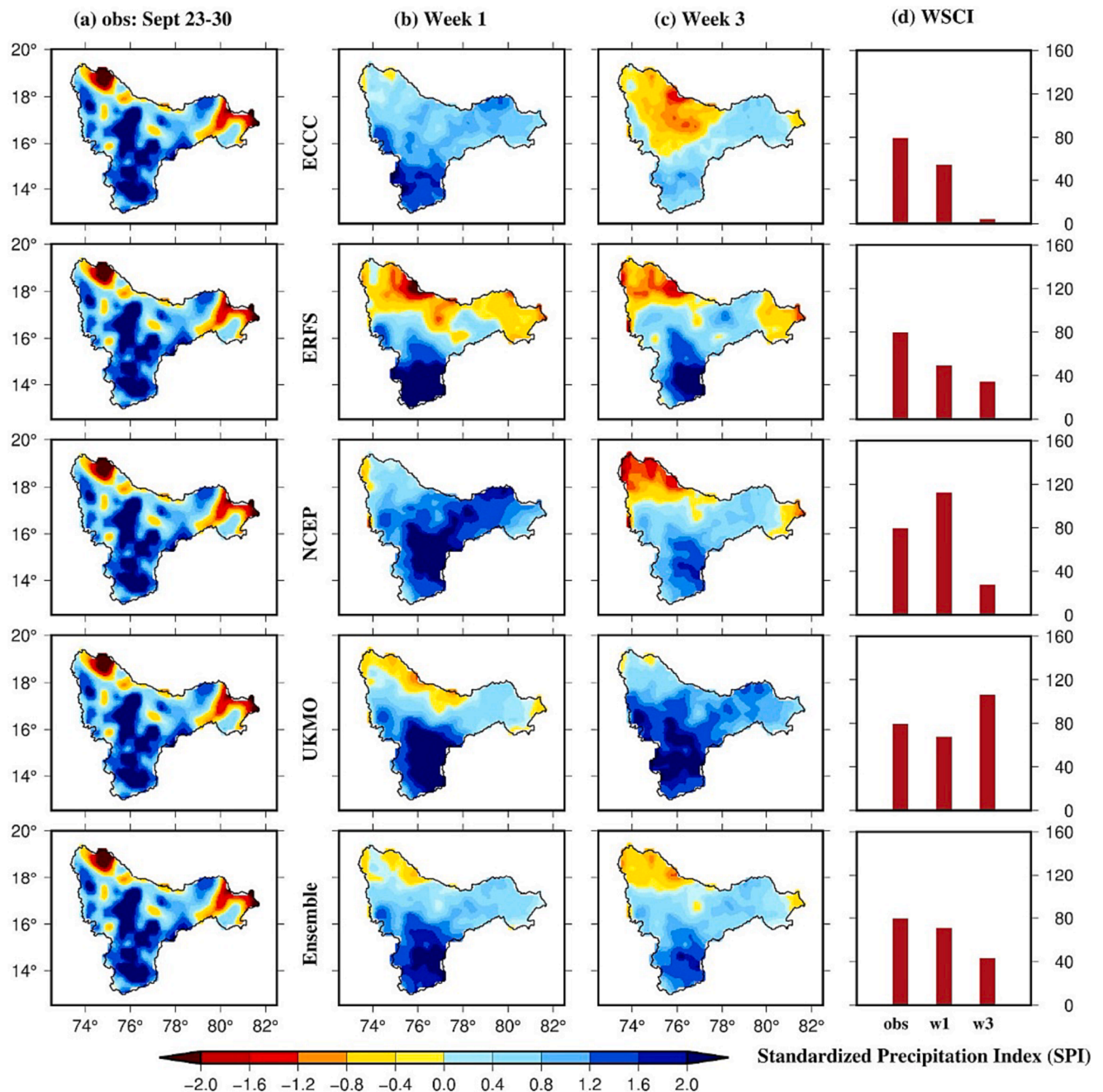


Fig. 10. (a–d) Comparison of September 2009 wet based on (a) observed SPI against S2S model initialized (b) one week, (c) three weeks and (d) five weeks before extreme wet event. (e) WSCI of wet event from observed data (obs) and from models initialised 1 week (w1) and 3 weeks (w3) before of the wet event respectively.

into a “ready-set-go” framework between seasonal and medium-range forecasts, by providing an early warning of an extreme event a few weeks in advance. However, we need more regional studies quantifying the benefits of S2S predictions of different extreme events to fully integrate these forecasts into a ready-set-go system to provide application-specific solutions. Future studies could explore bias correction for S2S forecasts as a method to enhance overall predictability. We also need to engage with stakeholders to further examine the utility of the S2S models in operational forecast.

Author credit

VM designed the study. IM conducted the analysis and wrote the first draft. Both authors contributed in the discussion.

CRediT authorship contribution statement

Iqura Malik: Data curation, Formal analysis, Methodology, Visualization, Writing – original draft. **Vimal Mishra:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

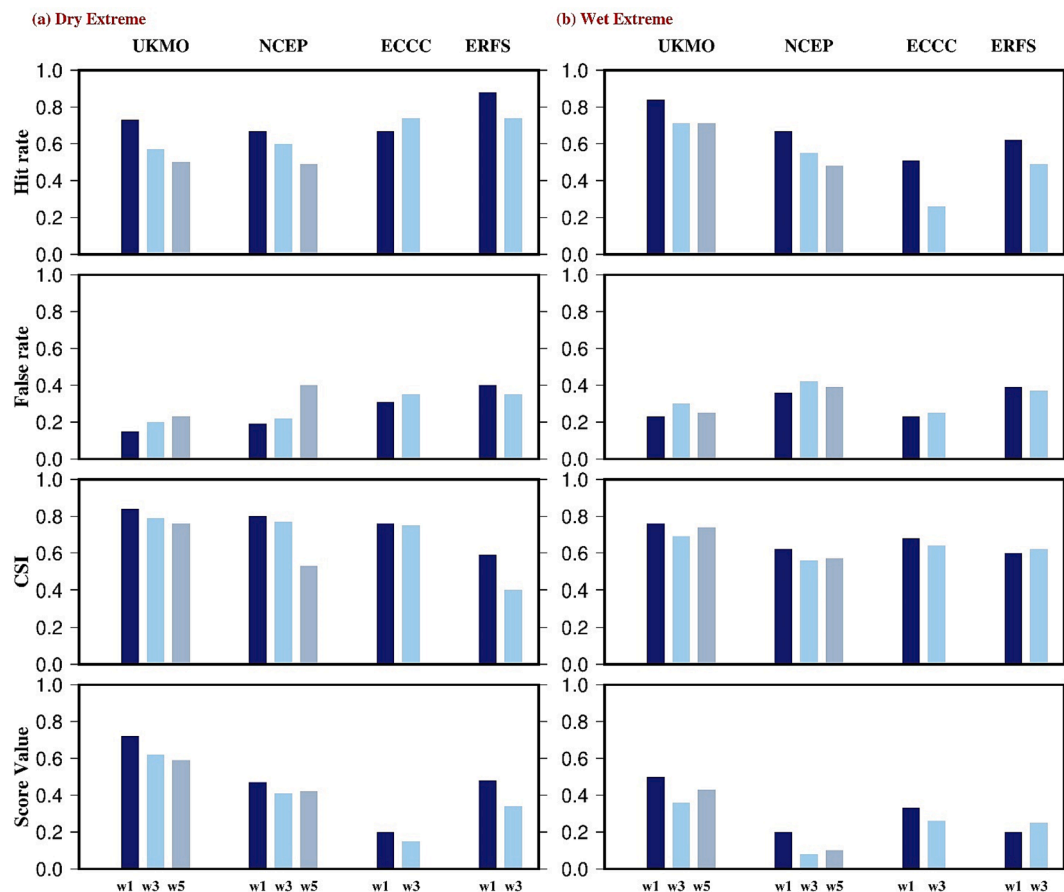


Fig. 11. Performance Evaluation of S2S models based on different categorical skill scores such as Hit rate, False rate, Critical Success Index (CSI) and Score Value for (a) dry extreme and, (b) wet extreme events computed during a common time frame (2003–2010) at different lead scale (week1, week2, week3 and week5) denoted by w1, w2, w3 and w5 respectively, based on the model's lead time availability.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cliser.2024.100457>.

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